

: Hugging Face Transformers:

A cat sat on the ————

mat
floor
chair
LED

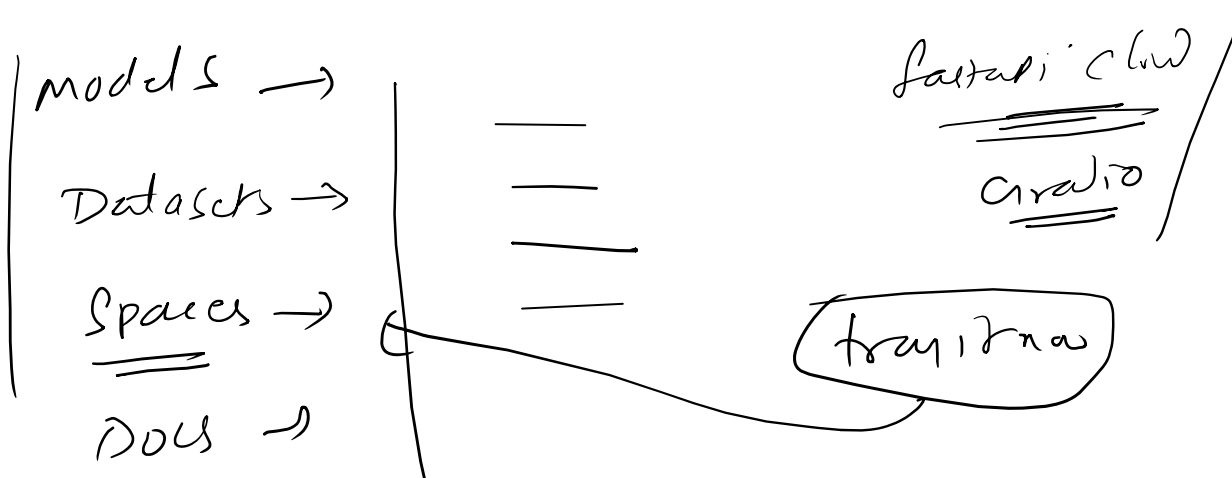
50%
25%

Hugging Face

Transformer → neural network architecture

Hugging face transformer → software library

Hugging Face Hub → platform / repository
of models, datasets,
demos, documents.

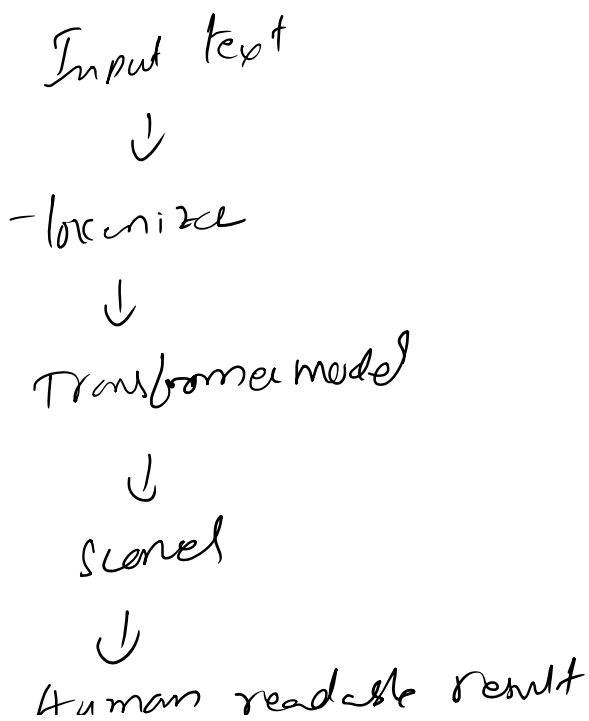


: pipeline: ① you say the task

Sentiment Analysis
question answering

Summary
Question Answering
translation
Summarization.

- ② It prepares the input
it turns your words into a
form a model can read
- ③ It runs the model
uses sensible default model
you don't have to choose one
- ④ It Answers
gives you readable text
sure.



↓
Human readable result

pipeline → it is a easy button

Raw text → "I love this movie"

↓

tokenizer

↓

[101, 102, ...]

↓

Model

↓

softmax

↓

positive = 99%

↓

(human readable format)

pipeline ("sentiment_analysis")

means

↓

"Give me a transformer configured for this type of job"

from transformers import pipeline

... = pipeline("text-classification", model="...")

Hyman

perplexity = Pipe In (4 top-sequences), model = "carry over"

Checkpoint:

student before studying

↓

Training

↓

student after training

Transformer architecture

↓

Training

↓

learned weights

save set of
learned parameters

BERT
+

learned weights
=

Checkpoint 1

I love transformers,

↓

problem

↓

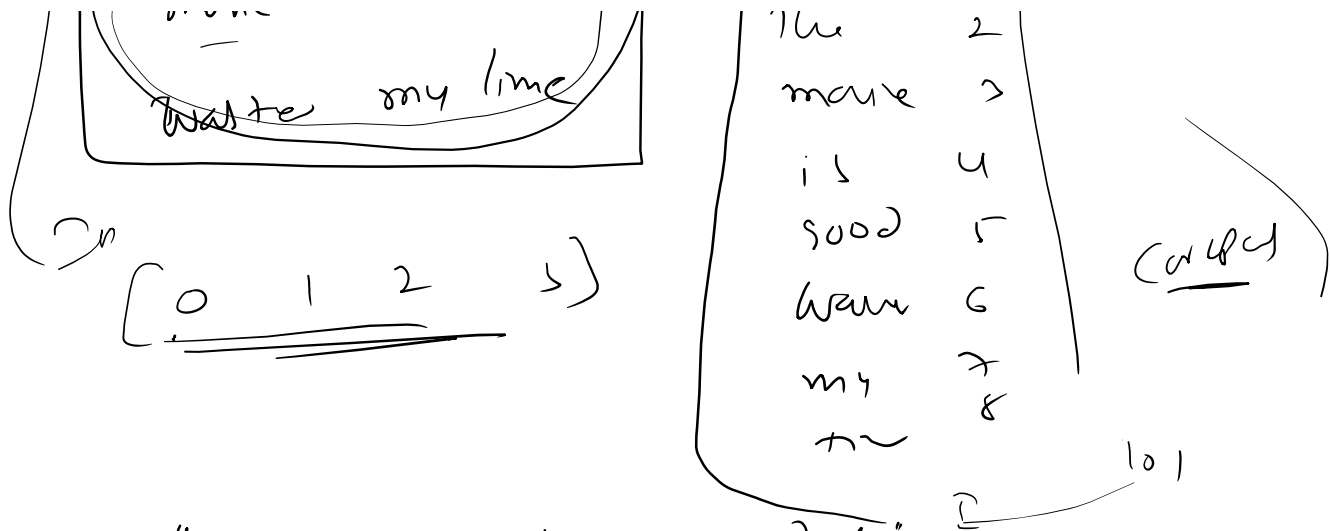
cross
entropy

How the model

det

I like the more
+
more it grows
my time

I	0
like	1
The	2
more	3



Arbitrary (A c) (ute) (ted)

WORD "A" "cute" "teddy bear" "is"

Sub-word

"A" "cute" "teddy" "bear" "is" "reading"

Dataset



tokenizer



model checkpoint



Train - Fine tuning



Evaluation



output

output
↓
Decode

- Human n= walk about,

playing

play
my

<UNK>

" A cat sat on the ? "

mat
floor
chair
bed

Sentence → Tokenisation → Token IDs → Embeddings

→ position Information → Self attention →

Feed-Forward neural network → transformer

layer → look → softmax → probability of

next token → "mat"

A cat sat on the

unselectable

un
believ
ance

A	0
cat	1
Sat	2

. . . cat on the

Set 2
on 3
the 4
not 5
flur 6
dos 7

A cat sat on the
0 1 2 3 4

cat = 1 sat = 2

$$\text{sat} = 2 \times \text{cat} - X$$

one hot encoding

0	0.0
1	0.0
2	0.0

$$A = [1, 0, 0, 0, 0]$$

create embedding

$$\text{cat} = [0, 1, 0, 0, 0]$$

$$\text{sat} = [0, 0, 1, 0, 0]$$

on —

the —

A simply list of learned numbers representing
to v-m.



$$\text{cat} = [0.8 \quad 0.2]$$

$$\text{dog} = [0.25 \quad 0.75]$$

$$\text{car} = [-0.8 \quad 0.5]$$

Real 11m.

$$A \rightarrow [0.1, 0.2]$$

$$\text{cat} \rightarrow [0.8, 0.2]$$

$$\text{sat} \rightarrow [0.5, 0.5]$$

$$\text{on} \rightarrow [0.2, 0.8]$$

$$\text{the} \rightarrow [0.3, 0.6]$$

$$\text{cat} \rightarrow$$

$$\text{the} \times \text{RE}$$

$$X = [0, 1, 0, 0, 0]$$

$$\text{tokens}$$

$$\text{IN}$$

$$\downarrow$$

$$\text{embedding}$$

$$= [0, 1, 0, 0, 0]$$

$$\begin{bmatrix} 0.1 & 0.2 \\ 0.8 & 0.2 \\ 0.5 & 0.5 \\ 0.2 & 0.8 \\ 0.3 & 0.6 \end{bmatrix}$$

$$= [0 + 0.8 + 0 + 0 + 0, 0 + 0.2 + 0 + 0 + 0] = [0.8, 0.2]$$

$$\text{Dog} \quad \text{sites} \quad \text{man}$$

$$\text{man} \quad \text{sites} \quad \text{dog}$$

Position Information

A position
cat position
sat position
on position
the position

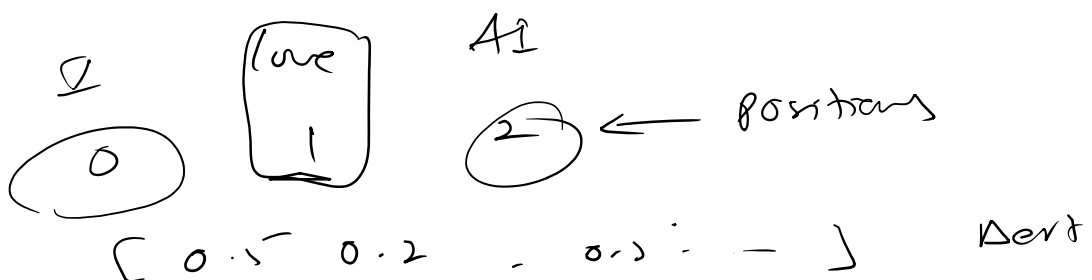
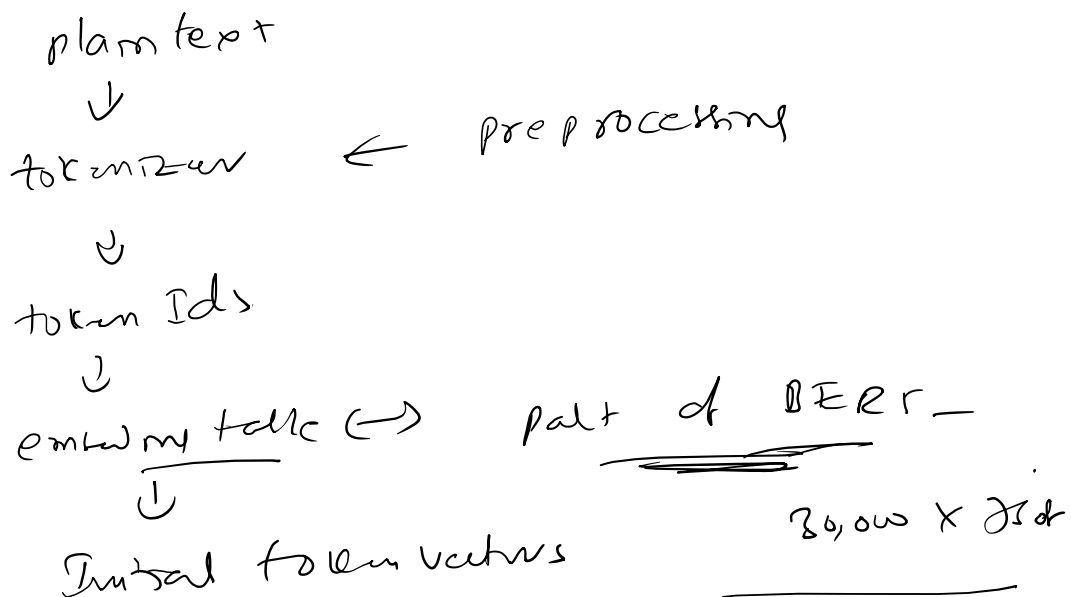
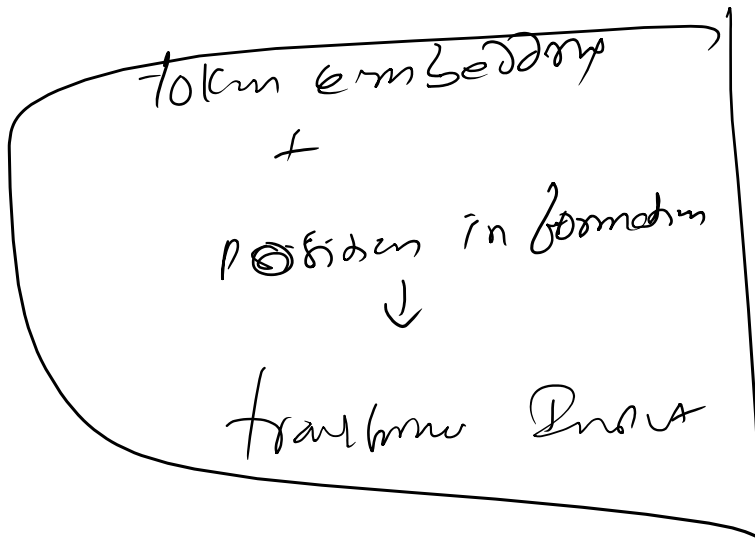
cat embedding

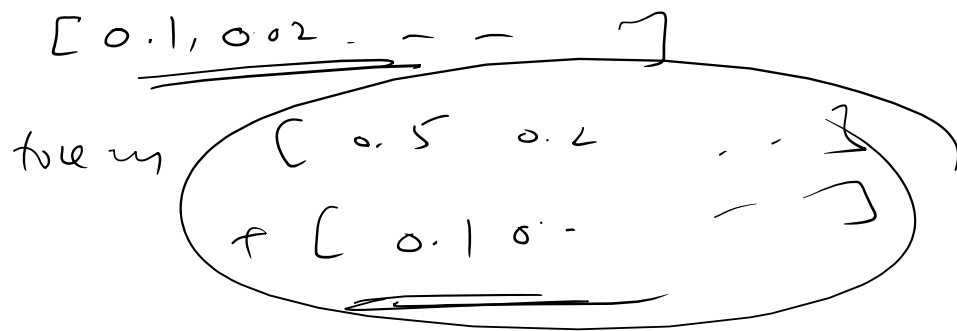
$$[0.8, 0.2] \quad d_k = 2$$

$$\text{pos} = [0.1, 0.0]$$

The process

$$\text{Embedding of position} = \underline{\underline{[0.5, 0.2] + [0.1, 0.3]}} \\ = \underline{\underline{[0.6, 0.5]}}$$





token $\rightarrow$ token_{id}, Embedding, learn + learn,
 Position Embedding

Attention:

A cat sat on the —

P_1	P_2	P_3
P_4	—	

A cat sat on the —

A

cat on

sat on the

on the

The useful

Q, K, V

... the ...

A	—	0.2
cat	—	1.1
set	—	1.2
<u>on</u>	—	<u>1.6</u>
<u>the</u>	—	<u>0.3</u>

set ON the

Query $\rightarrow$ what am I looking for

Q = Query

key $\rightarrow$ k what information does each row represent

value $\rightarrow$ v what useful information I actually take from a row

the = [0.3 0.6] w_Q, w_k, w_v

$$Q = X w_Q$$

$$K = X w_K$$

$$V = X w_V$$

... / $Q K^T$

$$\text{Attenuation} = \frac{\sum \left(\frac{Q_{ik}^2}{\sqrt{d_k}} \right) V}{\dots}$$

$$Q \times K$$

$$② \text{ divide by } \underline{\sqrt{d}}$$

$$③ \text{ get max}$$

$$④ \text{ multiply } V$$

$$Q = [1, 1]$$

$$A = [0.1 \ 0.1]$$

$$Q = [1, 1]$$

$$\text{cat} = [0.9 \ 0.2]$$

$$\text{sat} = [0.4 \ 0.5]$$

$$\text{on} = [0.6 \ 0.8]$$

$$\text{the} = [0.2 \ 0.5]$$

$$Q \times K = [1, 1] \times [0.8, 0.8]$$

$$= 0.8 + 0.8$$

$$= \underline{\underline{1.6}}$$

A cat sat on the

0.2 1.1 1.3 1.6 0.7

[_ _]

$$\sqrt{D_{11}} = \underline{\underline{1.414}}$$

A	cat	fat	on	th
0.2 / 1.414	1.1 / 1.414	1.3 / 1.414	1.6 / 1.414	0.7 / 1.1
0.14	0.78	0.92	1.13	0.49

$A \rightarrow 0.14$
 $cat \rightarrow 0.78$
 $fat \rightarrow 0.92$
 $on \rightarrow 1.13$
 $th \rightarrow 0.49$

$A \rightarrow 0.109$

$\underline{\underline{S_{ot} \max}}$ $cat \rightarrow 0.206$
 $fat \rightarrow 0.28$
 $on \rightarrow 0.283$
 $th \rightarrow 0.17$

$$e^{0.14} + e^{0.78} + e^{0.92} + e^{0.13} + e^{0.49}$$

$A = [\quad]$
 $\underline{\underline{cat}}$
 $\underline{\underline{fat}}$
 $\underline{\underline{on}}$
 $\underline{\underline{th}}$

$A \quad 0.109$
 $cat \quad 0.206$
 $fat \quad 0.28$
 $on \quad 0.283$
 $th \quad 0.1722$

th

1 []

Sentence